Precambrian Research 272 (2016) 1–17

Contents lists available at ScienceDirect

Precambrian Research

journal homepage: www.elsevier.com/locate/precamres

Reassessment of Aguapeí (Salto do Céu) paleomagnetic pole, Amazonian Craton and implications for Proterozoic supercontinents

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ARTICLE INFO

Article history:
Received 12 May 2015
Received in revised form 16 October 2015
Accepted 29 October 2015
Available online 7 November 2015

Keywords:
Amazonian Craton
Mesoproterozoic
Paleomagnetism
Salto do Céu Intrusive Suite
Nuna supercontinent
Rodinia supercontinent

ABSTRACT

The Aguapeí paleomagnetic pole obtained for mafic sills and dykes from Salto do Céu region at the western margin of the Amazonian Craton constrains its links with Baltica and Laurentia in Rodinian reconstructions. A new U–Pb age on baddeleyite at 1439 ± 4 Ma for the intrusives, contrasts strongly with a previous Ar–Ar age at 981 ± 2 Ma, with important consequences for paleogeographic reconstructions. We report new paleomagnetic and magnetic anisotropy results for sills from the Salto do Céu region and reassess the paleomagnetic data in view of the new geochronological age. A total of 155 samples were collected from thirteen new paleomagnetic sampling sites, five of them corresponding to sedimentary rocks located at the borders of the sills in an attempt to perform baked contact tests. After thermal and alternating field demagnetization, the sills provided a characteristic magnetic component at D_m = 208.2°, I_m = 68.5° (N = 8, α_{95} = 6.4°), with a corresponding paleomagnetic pole at 46.4°S; 277.0°E (A_{95} = 10.2°). Directions obtained in this study are similar to those reported previously for other mafic sills and dykes in the same region. A pole integrating the new results for the sills with those of the previous Aguapeí pole is situated at 56.0°S; 278.5°E (A_{95} = 7.9°). This new combined Salto do Céu pole supersedes the previous Aguapeí pole. Magnetic mineralogy studies, optical and electronic microscopy indicates PSD magnetite as the main magnetic carrier in these rocks. The baked contact tests were inconclusive, but the similarity between Salto do Céu pole and other high-quality poles with ages around 1420–1430 Ma suggests they carry a primary thermoremanence of that age. The Salto do Céu and other coeval poles are compatible with a connection between Amazonia and Baltica at the Mesoproterozoic in a paleogeographic configuration slightly different from SAMBA (*South America-Baltica*). At the same time, the new geochronological and paleomagnetic data imply that all paleogeographic interpretations for the position of Amazonia in Rodinia based on the previously published Aguapeí pole (and now renamed as Salto do Céu pole) must be revised.
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1. Introduction

The supercontinent cycle seems to have operated since Archean times, with at least three supercontinents formed since then, at 1.85–1.60 Ga (Columbia or Nuna), at 1.0–0.9 Ga (Rodinia) and at 0.28 Ga (Pangea) (Meert, 2012). Earlier Neoarchean continental assemblies have also been proposed such as Vaalbara, Superia and Sclavia (see Bradley, 2011, and references therein), but cannot be considered as supercontinents due to their limited sizes (Bleeker, 2003; Meert, 2012). A Paleoproterozoic supercontinent existed at about 1.85 Ga ago (Rogers, 1996; Rogers and Santosh, 2002; Meert, 2002; Meert and Torsvik, 2003; Cordani et al., 2003, 2009; Pesonen et al., 2003; Zhao et al., 2003, 2004; Evans, 2009; Johansson, 2009, 2014; Evans and Mitchell, 2011; Meert, 2012), although recent ideas are that it was not fully formed until 1.6 Ga ago (Evans, 2013; Pisarevsky et al., 2014; Pehrsson et al., 2015). Its rupture preceded the assembly of Rodinia at ca. 1.0–0.9 Ga (e.g., Weil et al., 1998; D'Agrella-Filho et al., 1998; Dalziel, 1997; Li et al., 2008). The name of the Paleoproterozoic supercontinent has been disputed, with some authors using Nuna and others Columbia (see discussion in

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http://dx.doi.org/10.1016/j.precamres.2015.10.021
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Meert, 2012; Evans, 2013). Here we adopt Nuna. Another fundamental question concerns the timing and mode of Nuna break-up and later Rodinia amalgamation (see Evans, 2013).

The Amazonian Craton is one of the greatest Precambrian cratons in the world and a key piece of Earth's paleogeography. Recently, some Proterozoic paleomagnetic poles were reported for the Amazonian Craton with important implications to Nuna (Bispo-Santos et al., 2008, 2012, 2014a; D'Agrella-Filho et al., 2012; Reis et al., 2013) and Rodinia (Tohver et al., 2002; D'Agrella-Filho et al., 2008; Elming et al., 2009a). The 1780 Ma Avanavero paleomagnetic pole from northern Guiana Shield (Amazonian Craton) published recently by Bispo-Santos et al. (2014a) supports a SAMBA-like connection (Johansson, 2009) where the southern boundary of Baltica faces the northwestern boundary of the Amazonia (see Pisarevsky et al., 2014 for alternative models). According to Bispo-Santos et al. (2014a) and also Evans (2013) and Evans and Mitchell (2011), Laurentia, Baltica, proto-Amazonian Craton and West Africa formed the nucleous of Nuna. The duration of this nucleous is still controversial. Some authors suggest it broke up soon after its amalgamation, at 1.78 Ga (e.g., D'Agrella-Filho et al., 2012), others suggest the break-up occurred later at about 1.5 Ga (e.g., Zhao et al., 2004; Rogers and Santosh, 2009), and others suggest that it was yet more long-lived (e.g., Vigneresse, 2005; Evans, 2013; Hou et al., 2008). Nuna has been characterized by a peculiar tectonic style, with decrease in subduction flux and subduction-related magmatism (Silver and Behn, 2008) and major emplacement of anorogenic rapakivi granites between 1.6 and 1.3 Ga (e.g. Hoffman, 1989; Anderson and Morrison, 1992; Åhäll and Connelly, 1998; Bettencourt et al., 1999; Karlstrom et al., 2001; Rämö et al., 2003; Vigneresse, 2005).

Concerning Rodinia, Tohver et al. (2002) proposed an oblique collision of the Amazonian Craton in the Texas area (the Grenvillian age Llano belt) of Laurentia at 1.2 Ga, based on the Nova Floresta pole from the Amazonian Craton. This idea was further reinforced by geological and isotopic data (Tohver et al., 2004a, 2004b, 2005a, 2005b, 2006), and new paleomagnetic data obtained for sedimentary rocks from the Fortuna Formation (the basal unit of the Aguapeí Group) dated at 1.15 Ga by U–Pb systematics on diagenetic xenotime overgrowths in zircon grains (D'Agrella-Filho et al., 2008), and for mafic sills and dykes dated at 0.98 Ga by ^{40}Ar–^{39}Ar on whole rock (Elming et al., 2009a). This model was also followed by Ibanez-Mejia et al. (2011) to explain the ca. 1.15 Ga Colombian-Oaxaquian fringing-arc accretion onto the northern margin of Amazonia and the final collision between Amazonia and Baltica at ca. 1.00–0.98 Ga along the Sveconorwegian orogen. Alternatively, using the opposite polarity of Amazonian poles and considering a long-lived Nuna, Evans (2013) suggested the Amazonia and Baltica performed clockwise rotations before their final collisions along the Grenvillian/Svecovergian/Sunsás orogens at about 1.0 Ga (model B in his Fig. 3) (see also Johansson, 2014).

An important piece of evidence for the first model is the paleomagnetic pole obtained by Elming et al. (2009a) for mafic sills and dykes that cut sedimentary rocks around Rio Branco town (Mato Grosso state), which are traditionally included in the intermediate unit (Vale da Promissão Formation) of the Aguapeí Group (Saes and Fragoso-Cesar, 1994). These sills and dykes belong to the Salto do Céu Intrusive Suíte (Araújo-Ruiz et al., 2005, 2007), but Elming et al. (2009a) called them Aguapeí (hereafter we will use the original Salto do Céu name). Paleomagnetic analysis revealed southwest (northeast) directions with moderate to steep downward (upward) inclinations for samples from ten sites. A mean direction $D_m = 11.3^\circ$; $I_m = -57.9^\circ$ ($\alpha_{95} = 8.1^\circ$; $K = 37$) was calculated which yielded a paleomagnetic pole at 64.3°S; 271.0°E ($A_{95} = 9.2^\circ$) (Elming et al., 2009a). The limited number of sites and the absence of a baked contact test for the Salto do Céu intrusives motivated a resampling of this area. Meanwhile, a new U–Pb age on baddeleyite of 1439 ± 4 Ma was obtained (Teixeira et al., 2015) that contradicts the previous Ar–Ar age on whole rock of 981 ± 2 Ma (Elming et al., 2009a). In this work, paleomagnetic analyses were performed for other mafic sills in the same region in order to improve statistics of the previously published paleomagnetic pole. At five outcrops, good exposures of sills in contact with host sedimentary rock allowed sampling for baked contact tests. We integrate these new results into a more robust paleomagnetic pole and discuss the tectonic implications to Nuna in view of the new U–Pb in baddeleyite age.

2. Geological setting

The Amazonian Craton is exposed in the Guiana and Brazil-Central (or Guaporé) shields, which are bisected by sedimentary rocks of the Amazon basin (Fig. 1a). According to Cordani and Teixeira (2007) its Precambrian evolution is marked by Archean primitive nuclei whose collision originated the Maroni-Itacaiunas Province (2.25–2.05 Ga), followed by successively younger accretionary magmatic events with minor or major juvenile crust involvement along its southwestern border. These accretionary events were divided in geochronological provinces: Ventuari-Tapajós Province (1.98–1.81 Ga), Rio Negro-Juruena Province (1.78–1.60 Ga), Rondonian-San Ignacio Province (1.60–1.30 Ga) and Sunsás-Aguapeí Province (1.20–0.95 Ga). The latter forms the southwestern tip of the Amazonian Craton (Fig. 1a).

The Aguapeí Group, whose type area is situated at the Brazil/Bolivia border near Vila Bela (Fig. 1b), represents a basin predominantly composed of clastic sediments typical of a continental shelf, showing transgressive–regressive phases related to a marine environment (Menezes et al., 1991; Saes et al., 1992). Saes and Fragoso-Cesar (1994) subdivided this unit from base to top into three formations: Fortuna, Vale da Promissão and Morro Cristalino. According to these authors the depositional history of the Aguapeí Group can be described by three stages: rift stage, marine stage with subsidence, and inversion stage with reworking. The Aguapeí Group is correlated with the Huanchaca Group that is exposed in the southern extremity of the Paraguá Terrain in Bolivia (Saes and Fragoso-Cesar, 1994, Fig. 1b). U–Pb datings of zircon grains from Fortuna Formation at São Vicente gold mine (Mato Grosso state) yielded a youngest age of 1165 ± 27 Ma for one of the grains which sets the maximum deposition age of this formation (Santos et al., 2001). U–Pb SHRIMP analysis on authigenic xenotime rims on detrital zircons yielded an age of 1149 ± 7 Ma for the diagenisis of red beds from the Fortuna Formation (D'Agrella-Filho et al., 2008). These data constrain the age of the basal formation of the Aguapeí Group to between 1165 ± 27 Ma and 1149 ± 7 Ma.

The Salto do Céu Intrusive Suite is represented by a set of mafic sills and dykes exposed in the Rio Branco, Salto do Céu and Vila Progresso regions (Araújo-Ruiz et al., 2005, 2007). These diabase rocks intrude undeformed siliciclastic sedimentary rocks (Fig. 1b). These rocks have been correlated to the Vale da Promissão Formation of the Aguapeí Group, mostly on the basis of K–Ar ages for the sills and dykes (on plagioclase and whole rock) that vary between 1015 ± 17 and 875 ± 21 Ma (Barros et al., 1982; Ruiz, 1992, 2005). ^{40}Ar–^{39}Ar dating on whole rock yielded a plateau age at 981 ± 2 Ma for the same rocks (Elming et al., 2009a). Recently, however, baddeleyites separated from a sill at Salto do Céu region yielded a U–Pb concordia age of 1439 ± 4 Ma, which is interpreted as the crystallization age of the sill (Teixeira et al., 2015). This age implies that the sedimentary rocks intruded by the sills and dykes are older than 1440 Ma and did not belong to the intermediate unit of the Aguapeí Group, as previously believed. U–Pb determinations on 100 detrital zircons separated from one sample from sediments in the same Salto do Céu region yielded age populations ranging from 2515 Ma to 1544 Ma which set a maximum age for the sedimentation at 1544 Ma (Geraldes et al., 2014). Another evidence for an older age for these sedimentary rocks is the occurrence of them as

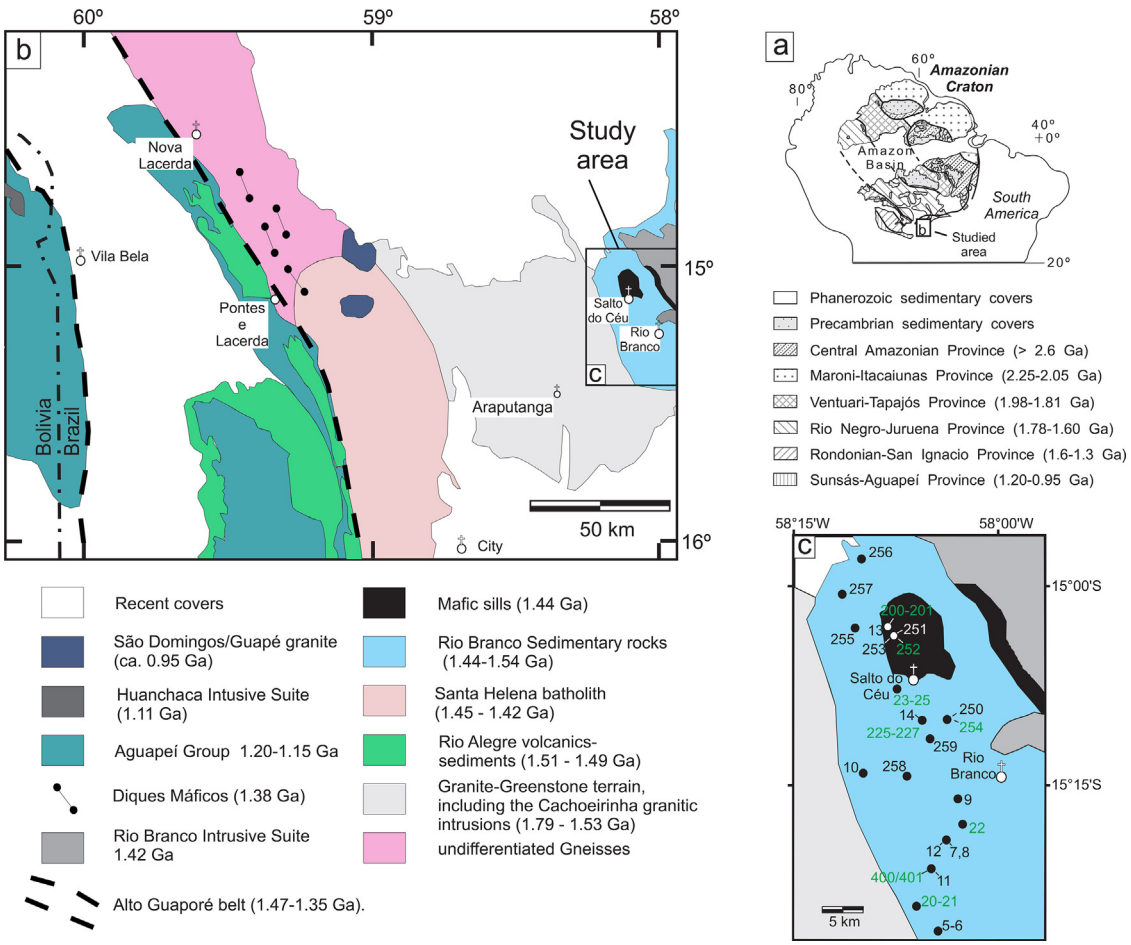


Fig. 1. (a) Main tectonic provinces of the Amazonian Craton (adapted from Cordani and Teixeira, 2007; Bettencourt et al., 2010); (b) Simplified geological map of the studied area (adapted from Geraldes et al., 2001; Ruiz, 2005); (c) Geographical positions of the sampling sites (full circles). Numbers in black/white (green) represent mafic sills (sedimentary rocks) sites (see Tables 1 and 2). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

xenoliths in igneous rocks from the anorogenic Rio Branco mafic–felsic intrusive suite (Ruiz, 2005), for which U–Pb ages of 1471 ± 8 Ma and 1427 ± 10 Ma on zircons were obtained for a gabbro and a granophyre, respectively (Geraldes et al., 2001). This unit is coeval with the Santa Helena batholith (1.45–1.42 Ga – Fig. 1b), a product of one of the accretionary arcs generated during the Paraguá block collision at the southwestern margin of the Amazonian Craton (see review of Bettencourt et al., 2010).

The Salto do Céu sills and dykes vary from 1 to 20 m in thickness. They consist of dark-gray to black, medium to fine-grained (rarely porphyritic) equigranular massive mesocratic rocks, with textures varying from micro-gabbros to diabases (Araújo-Ruiz et al., 2007; Araújo et al., 2009). The rock is composed of plagioclase, pyroxene, hornblende, biotite, potassic feldspar, and quartz. Zircon, titanite, magnetite, ilmenite, and pyrite are accessory minerals (Araújo-Ruiz et al., 2007; Araújo et al., 2009).

3. Sampling

Standard paleomagnetic cylindrical samples (diameter of ~2.54 cm) were collected with a portable gasoline-powered rock drill and oriented by both magnetic and solar compasses. Sixty-seven cylindrical cores were sampled from 8 sites from unmetamorphosed sills (Table 1). At five sites 87 cylindrical cores and one oriented hand sample from the adjacent sedimentary rocks were sampled at different distances from the sills for baked contact tests. From three of these sills, paleomagnetic results were already presented by Elming et al. (2009a) and correspond to their sites 11, 13 and 14.

Site 13 corresponds to an 8–10 m thick sill with the sedimentary host rock well exposed above it. Fifteen cylindrical cores were sampled from the sedimentary rocks (AZ200-A to AZ200-H and AZ200-S to AZ200-Z; there is no letter W – Table 2) from the contact (AZ200-S) up to 2.8 m above it (AZ200-Z). Ten to twelve meters above sample AZ200-Z, another 11 cores (AZ201-A to AZ201-K) were collected up to 5.1 m upsection. These samples represent the unbaked sediments relatively far from the contact.

Site 14 is a 15–20 m thick sill. At this site, sedimentary rocks at three levels above the sill were sampled which correspond to sites AZ225 (samples A to H), AZ226 (A to G) and AZ227 (A to O) totalizing 29 cylindrical cores. Site AZ255 was sampled from contact up to 30 cm; samples from site AZ226 were collected at stratigraphic height of 40 cm up to 1.15 m above the contact in an equivalent section but in another part of the outcrop; site AZ227 was sampled about 20 m above the contact over a ~2 m thick profile.

Site 11 was not well exposed hence the stratigraphic position of sedimentary rocks relative to the sill could not be established precisely. Samples comprise standard cylinders (AZ400-A to D) and a hand sample (AZ401). The reason why we have attempted to do a baked contact test in this sill is that it is the only one to yield northeastern directions with negative inclination. All other four sills where the contact with sedimentary rocks could be sampled yielded southwestern, steep positive inclinations directions (see Section 5).

Table 1
Paleomagnetic data for the Salto do Céu Intrusives.

Site	Glat/Glong	Site mean directions						VGP	
	(°S/°E)	*n*	Dec. (°)	Inc. (°)	α_{95} (°)	*K*	*N*, *R*	Plat. (°N)	Plong. (°E)
5[a]	15.433/301.925	6	184.8	45.9	5.6	144	*R*	−77.3	282.1
6[a]	15.434/301.925	6	205.7	68.9	12.0	30	*R*	−47.8	278.7
7[a]	15.320/301.937	6	21.8	−53.6	3.1	482	*N*	62.8	79.7
8[a]	15.317/301.936	6	17.1	−51.4	3.9	291	*N*	67.1	82.1
9[a]	15.268/301.949	8	22.0	−50.5	4.5	153	*N*	64.3	74.3
10[a]	15.234/301.836	8	143.3	49.2	12.6	20	*R*	−53.2	1.6
11[a]	15.354/301.916	4	2.8	−62.8	8.6	115	*N*	61.0	117.8
12[a]	15.316/301.936	7	11.6	−55.0	3.3	333	*N*	67.3	96.9
13[a]	15.054/301.862	9	213.8	65.2	2.1	131	*R*	−47.4	268.0
14[a]	15.169/301.906	23	202.5	61.8	2.1	205	*R*	−56.2	271.7
AZ250[b]	15.183/301.936	12	196.2	66.0	4.0	120	*R*	−54.3	283.4
AZ251[b]	15.064/301.867	4	242.0	72.9	9.9	88	*R*	−27.3	270.5
AZ253[b]	15.064/301.866	9	214.3	59.3	1.9	704	*R*	−51.0	258.6
AZ255[b]	15.054/301.823	8	200.0	65.5	2.0	779	*R*	−52.3	280.3
AZ256[b]	14.968/301.832	8	198.3	73.8	7.2	61	*R*	−43.2	289.3
AZ257[b]	15.014/301.810	13	205.3	60.0	4.4	90	*R*	−56.1	266.5
AZ258[b]	15.239/301.886	15	238.0	70.8	4.6	71	*R*	−30.5	267.6
AZ259[b]	15.193/301.917	9	178.1	71.1	2.5	414	*R*	−49.6	303.6
Mean *N*		5	15.8	−54.8	6.2	152	*N*	65.5	89.5
								A_{95} = 7.8°	*K* = 98
Mean *R*		13	198.3	66.0	7.2	34	*R*	−52.2	280.7
								A_{95} = 10.3°	*K* = 17
Mean *N* + *R*		18	197.4	62.9	5.7	38	*R*, *N*	−56.0	278.5
								A_{95} = 7.9°	*K* = 20

Glat/Glong – geographic latitude and longitude of the sampled sites; *n* – number of specimens used in the mean; Dec. – declination; Inc. – inclination; α_{95} – radius of the 95% cone of confidence; *K* – Fisher precision parameter; VGP – virtual geomagnetic pole; Plat. – pole latitude; Plong. – pole longitude.

[a] From Elming et al. (2009).

[b] This work.

The last two sites where baked contact tests were attempted (AZ252 and AZ254) correspond to sedimentary rocks intruded by sills (AZ251 and AZ250, respectively) recently discovered and not sampled by Elming et al. (2009a). Unfortunately the thickness of these sills could not be measured. In the vicinity of sill AZ251, 25 cylindrical cores of sedimentary rocks were sampled (site AZ252) from the contact (samples AZ252-A and AZ252-B) up to ~9.5 m above it (samples AZ252-W to AZ252-Z). In the vicinity of sill AZ250 only three cylindrical cores were sampled (site AZ254) at the contact with the sill (sample AZ254-A), at 10 cm from the contact (AZ254-C), and 2 m away from the contact (sample AZ254-B).

4. Analytic methods

Cylindrical cores were cut into 2.2 cm-high specimens and submitted to conventional stepwise thermal and alternating field (AF) demagnetization to isolate the characteristic remanent magnetization (ChRM) component carried by the samples. Normally, steps of 2.5 mT (up to 15 mT) and 5 mT (15–100 mT) were employed for AF demagnetization using a tumbler Molspin AF demagnetizer or an automated three-axis AF demagnetizer coupled to a 2G-Enterprises cryogenic magnetometer. Steps of 50 °C (100–500 °C), and 20 °C (500–600 °C) were used for the thermal demagnetization in a Magnetic Measurements TD-48 oven. Measurements of remanent magnetization were carried out using a fluxgate spinner Molspin magnetometer or a 2G-Enterprises cryogenic magnetometer. Both apparatuses are housed in a shielded room with an internal field smaller than 500 nT. Magnetic components were identified and calculated using orthogonal projections (Zijderveld, 1967) and principal component analysis (Kirshvink, 1980). At least four demagnetization steps were used to calculate vectors from least-squares fits, and only samples with mean angular deviation (MAD) smaller than 8° were retained. Fisher's (1953) statistics was used to calculate site mean directions and paleomagnetic poles.

Magnetic mineralogy was investigated performing hysteresis curves with a MicroMag-VSM, Model 3900 (Princeton Measurements Corporation), by the acquisition of isothermal remanent magnetization (IRM) in a MicroMag-VSM, Model 3900 (Princeton Measurements Corporation) or in a Magnetic Measurements pulse magnetizer, and using thermomagnetic curves performed with a CS3 furnace coupled to a Kappabridge KLY-4S instrument (Agico, Brno). Anisotropy of low-field magnetic susceptibility (AMS) was measured to determine the magnetic fabric of the rocks using the same KLY-4S. Mean AMS eigenvectors (maximum susceptibility, K_1, intermediate susceptibility, K_2, and minimum susceptibility, K_3) and 95% confidence regions were calculated using the bootstrap method of Constable and Tauxe (1990). All magnetic measurements were performed in the Laboratory of Paleomagnetism of the University of São Paulo.

Twenty-two thin and polished sections were microscopically investigated under transmitted and reflected light, respectively. Scanning electron microscopy (SEM) analyses were also performed to better constrain the mineralogy and textures of the fine-grained rocks.

5. Results

5.1. Petrography

Thin and polished sections from both diabase sills and host sedimentary rocks were investigated in order to constrain the main petrographic characteristics and magnetic mineralogy of studied samples. The petrographic descriptions allowed the distinction of different lithotypes in both groups (diabase sills and sedimentary rocks) and variable alteration degrees.

Less altered diabase samples are mostly medium- to fine-grained rocks, composed essentially of calcic plagioclase and clinopyroxene, with some samples containing olivine. They usually display well-defined subophitic to intergranular textures (Fig. 2a). The opaque minerals include magnetite, ilmenite, pyrite and minor chalcopyrite. Magnetite and ilmenite are euhedral to subeuhedral and pyrite occurs usually as an interstitial phase (Fig. 2a and b).

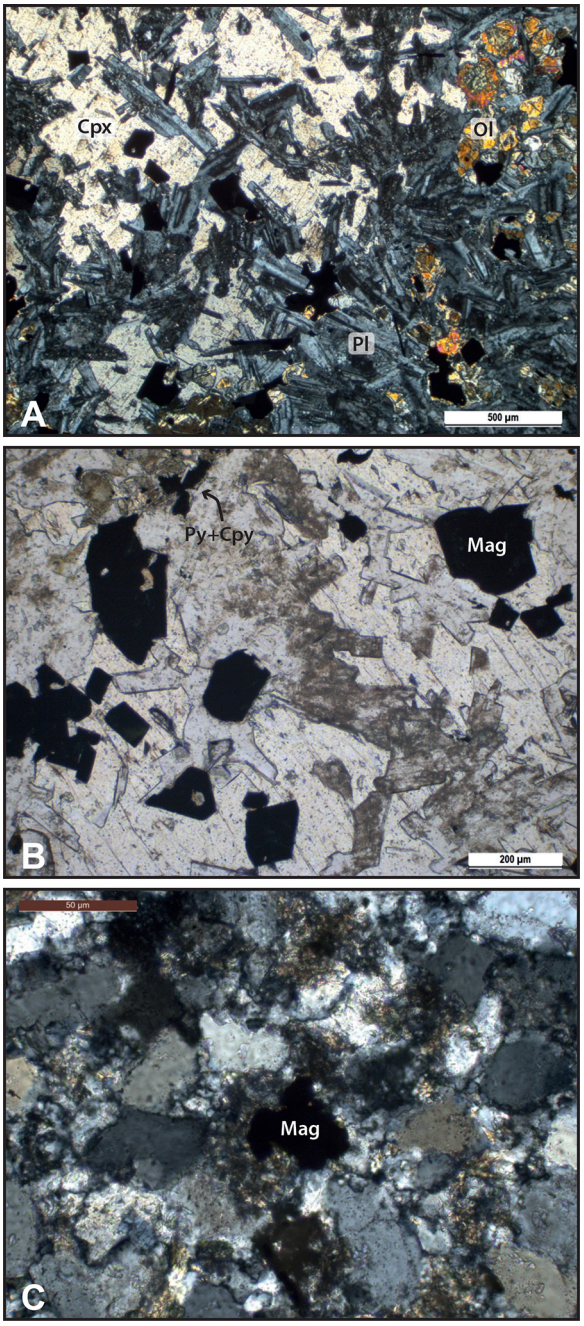


Fig. 2. Main textures and mineralogy of sills and sedimentary rocks from the studied area. (a) Photomicrography of medium- to fine-grained subophitic diabase, mainly composed of calcic plagioclase (Pl), clinopyroxene (Cpx), olivine (Ol), magnetite and ilmenite. Note the euhedral forms of opaque minerals (magnetite has cubic habit and ilmenite has tabular to acicular forms). (b) Parallel zoom in the same sample from (a) highlighting the euhedral forms of magnetite and interstitial grains of pyrite (Py) and chalcopyrite (Cpy). Note the presence of saussuritization of plagioclase (very fine-grained minerals giving a dirty aspect for the plagioclase). (c) Photomicrography of fine-grained sandstones showing compaction features of concave–convex contacts between quartz grains and post-compaction euhedral magnetite grain. Note the presence of very fine-grained unoriented white mica and chlorite associated with contact metamorphism. All photomicrographies were taken under transmitted light. The images (a) and (c) are in cross-polarized light, whereas the image (b) is in plane-polarized light.

Alteration is characterized mainly by discrete saussuritization of plagioclase and local replacement of clinopyroxene by amphibole and/or chlorite. In more altered samples, martitization of magnetite is well developed and seems to be associated with late-magmatic fluids rather than later metamorphic events.

The host sedimentary rocks comprise three lithotypes: (i) fine-grained arkosic sandstones; (ii) arenaceous siltstones; and (iii) siltstones. Most samples show evidence of low-grade metamorphism, frequently associated with crystallization of sericite and chlorite. Fine-grained arkosic sandstones are well compacted, showing framework of concave–convex contacts. They are composed essentially of quartz, minor plagioclase and subordinate detrital mica. Accessory minerals include zircon, apatite, magnetite, martite and pyrite. Arenaceous siltstones and siltstones have angular to subrounded quartz grains in a groundmass composed of very fine sericite and undifferentiated clay minerals. In the fine-grained rocks, opaque minerals include magnetite, hematite and pyrite. In general, the magnetic mineralogy of host sedimentary rocks includes euhedral magnetite crystals (locally martitized) and minor euhedral hematite. Detrital magnetite grains occur subordinately. Euhedral magnetite have grown over the compacted framework of the arkosic sandstone (Fig. 2c), indicating post-depositional and post-compaction crystallization. The iron oxides are associated with sericite and chlorite, possibly as a result of contact metamorphism from sill intrusions.

5.2. Magnetic mineralogy

Igneous rocks presented almost reversible thermomagnetic curves, most of them with a characteristic Verwey transition at low temperatures and a pronounced Hopkinson peak at high temperatures, just below a strong drop in susceptibility at around 580 °C, suggesting SD/PSD magnetite as the main magnetic carrier in the sills (Fig. 3a–c).

Sedimentary rocks presented irreversible curves typical of rocks with predominance of paramagnetic minerals, where a new magnetic mineral (probably magnetite) was formed during heating (Fig. 3d).

Isothermal remanent magnetization (IRM) acquisition curves were also obtained for sills (Fig. 4a) and sedimentary rocks (Fig. 4b). The sills show saturation fields normally smaller than 300 mT, which are typical of magnetite. For sedimentary rocks, IRM curves suggest that magnetite is also the predominant magnetic mineral in almost all samples, although a small amount of hematite is also present in samples AZ200-A2, AZ200-F1, AZ200-S3, AZ201-D3 AZ225-A5 and AZ226-B4 (Fig. 4b). For some samples, however, hematite is responsible for up to 80% of induced magnetization (e.g., AZ201-H1 and AZ201-J3). These samples are usually not saturated in fields of up to 2.5 T (Fig. 4b).

Hysteresis loops were performed for one sample from each site, also comprising sills and sedimentary rocks. The low magnetic intensity associated with sedimentary rocks resulted in very noisy curves, although the presence of magnetite is clear in the example of Fig. 5b. Typical hysteresis curves obtained for the mafic sills are shown in Fig. 5c–e. They give bulk coercive force (Hc), saturation magnetization (Ms), and saturation isothermal remanence (Mrs), after subtraction of the paramagnetic contribution from the high field portion of the curve. Coercivity of remanence (Hcr) was obtained using the same MicroMag VSM instrument. Coercivities (Hc) for the sills vary from 6 to 18 mT, which is typical of magnetite. Fig. 5a shows the Day's diagram (Day et al., 1977), which plots the Mrs/Ms and Hcr/Hc ratios. The SD, PSD and MD fields and the SD-MD and SD-SP (10%) theoretical mixing lines are those proposed by Dunlop (2002a), Dunlop (2002b). The samples fall along a trend parallel to the SD-MD magnetite mixing curves.

5.3. Anisotropy of magnetic susceptibility (AMS)

Magnetic anisotropy degree ($P = k_1/k_3$) values for all samples are lower than 7.5% with a peak in distribution of 3% (Fig. 6a), which is compatible with undeformed and unmetamorphosed intrusive

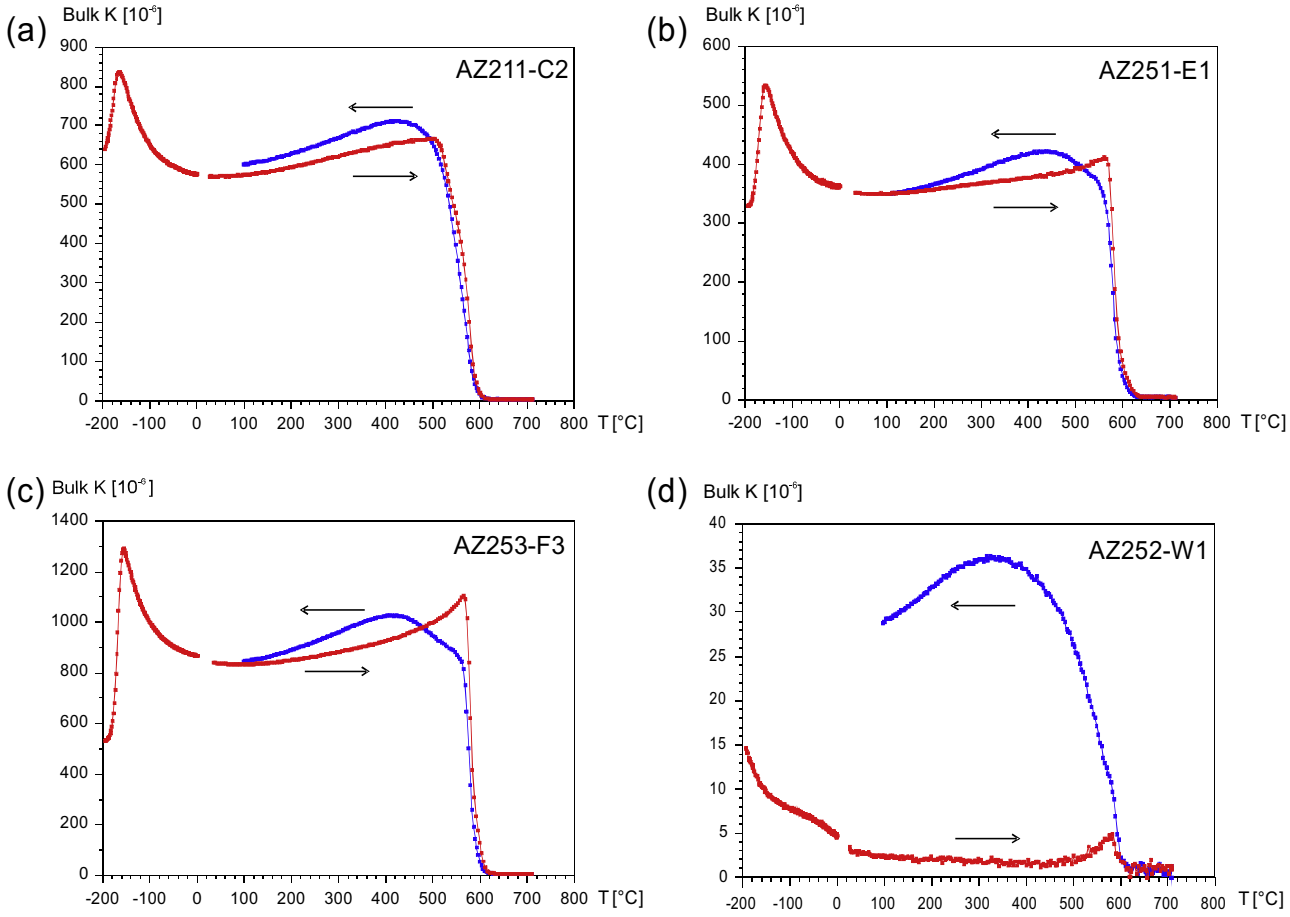


Fig. 3. Thermomagnetic (bulk susceptibility (K) versus temperature) curves for selected samples: sills rocks (a–c), sedimentary rock (d). Arrows indicate the heating (in red) and cooling (in blue) curves. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

and sedimentary rocks. A clear difference between mean magnetic susceptibility (K_m) is observed for sedimentary and intrusive rocks (Fig. 6b). While for sedimentary rocks K_m is between 1×10^{-4} and 10×10^{-4} (SI), with a median of 4×10^{-4} (SI), the sills yielded values between 10×10^{-4} and 316×10^{-4} (SI), with a median of 160×10^{-4} (SI).

Magnetic fabrics for sedimentary and mafic rocks are shown in Fig. 6c. Similar planar fabrics were obtained for all investigated sites, where minimum susceptibility axis (k_3) is approximately vertical, and intermediate (k_2) and maximum (k_1) susceptibility axes are unevenly distributed around the border of the stereoplot (Fig. 6c). This feature is compatible with flat lying sills and sedimentary rocks, and also, means that probably no tectonic events affected the rocks after their emplacement. Most values of Jelinek (1981) parameter (T) are between 0 and 1 (Fig. 6d), which implies a predominance of oblate ellipsoids, also compatible with the primary fabrics expected for the analyzed rock-types.

5.4. Paleomagnetism

Fig. 7 shows examples of AF and thermal demagnetization for specimens belonging to the mafic sills. Both treatments revealed

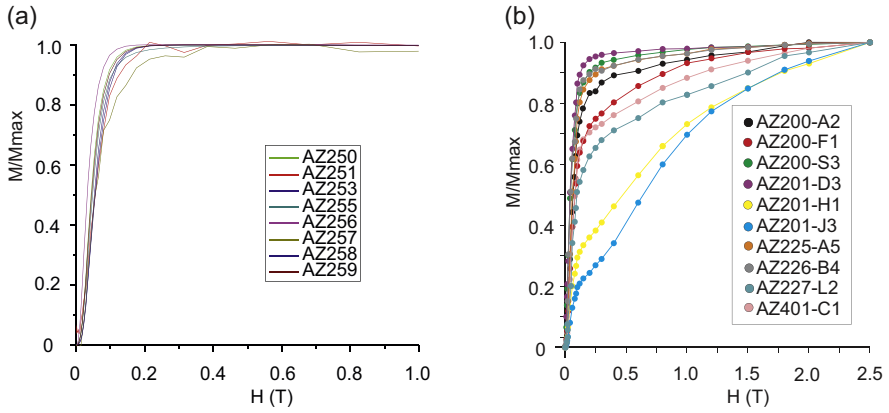


Fig. 4. Normalized isothermal remanent magnetization (IRM – M/Mmax) versus applied field (H) curves for samples from the sills (a) and sedimentary rocks (b).

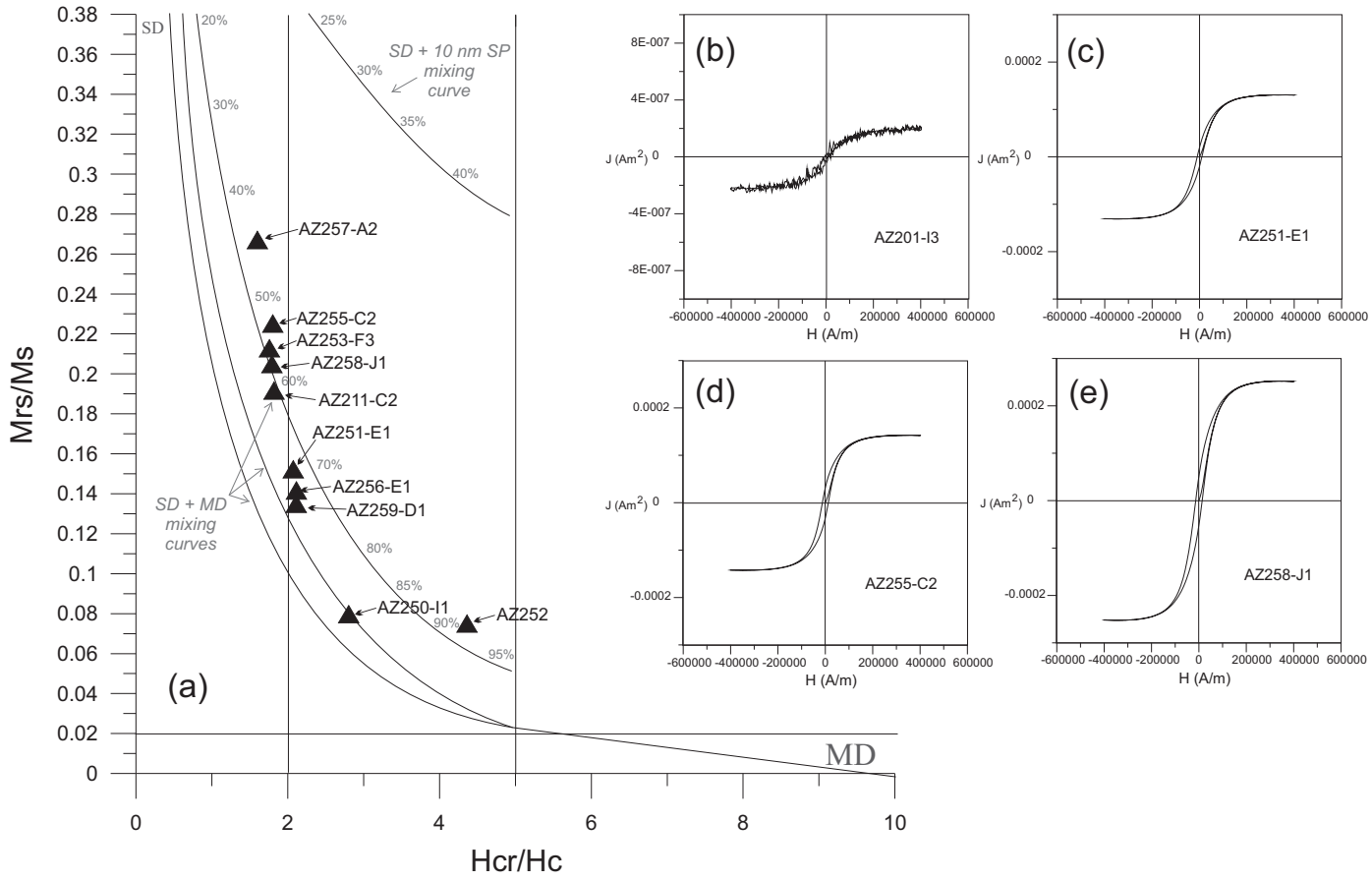


Fig. 5. (a) Day's diagram: Mrs/Mr versus Hcr/Hc ratios (Day et al., 1977) obtained for the sills. The limits between SD, PSD and MD fields and the SD/MD and SD/SP (with SP grain sizes of 10 nm) mixing curves are those proposed by Dunlop (2002a, 2002b). Examples of hysteresis curves are also shown for samples from the sedimentary rocks (b), and from sills (c and d).

southwestern, steep downward directions for all analyzed sites, which are similar to the previous results reported by Elming et al. (2009a) (see below). Note however that the new directions are mostly concentrated at high inclination pointing to the southwest. Thus the average obtained with the addition of the new directions is significantly different from the previous ChRM reported by Elming et al. (2009a). That is because the previous study was based on much less results and the two sites with lower inclination (sites 5 and 10; Table 1) had a bigger influence in the mean. The larger dataset reported here likely helps to better account for the secular variation and provides a more robust paleomagnetic pole. The characteristic remanent magnetization (ChRM) is carried by magnetite normally with high coercivity and high-unblocking temperatures, and represents the last component carried by the rock (Fig. 7).

Sedimentary rocks collected at different distances from their respective sills (sites 13, 14, 251, and 250, see Table 2) almost always presented a similar southwest, downward inclination direction (Figs. 8 and 9). Normally, this ChRM direction is carried by Ti-poor titanomagnetite as can be seen by the strong decay around 550–580 °C during thermal demagnetization (Fig. 9). For several specimens, a second high-coercivity (>100 mT) north/northwest or southwest direction with low (<30–35°) inclination is evident after AF treatment (Fig. 8). This high-coercivity component is not far from recent normal or reversed geomagnetic fields. So, the best explanation is that it is probably carried by secondary hematite formed by chemical oxidation of magnetite in recent times as suggested by the petrographic evidence for martitization in several samples. Unfortunately, the thermal treatment did not help us to better characterize this component (Fig. 9).

For site AZ400/401 it was possible to isolate a northeast, upward inclination direction (Fig. 10a) for ten analyzed specimens (Table 2), which is similar to the ChRM direction of the adjacent sill (site 11 of Elming et al., 2009a, see above). Specimens from the hand sample yielded another component, which correspond to a very stable and steep upward southeast direction (MAD = 0.1) defined between 100 and 350 °C (mean for 4 specimens: Dec. = 133.8; Inc. = −78.1, α_{95} = 0.4°, K = 64,058). At higher temperatures these samples showed a less stable northeast component similar to that from the sill itself (site 11 in Table 1). The southeast component is carried by pyrrhotite produced at some undefined time (maybe at earliest Cambrian time, due to hydrothermal activity in the adjacent Paraguay belt?) after sediment deposition.

6. Discussion

6.1. New paleomagnetic pole for Salto do Céu sills

A summary of the paleomagnetic data for sills and sedimentary rocks are presented in Tables 1 and 2, respectively. Fig. 11a–c shows site mean directions for the sills: Fig. 11a shows the mean directions for the ten sites analyzed by Elming et al. (2009a), Fig. 11b includes mean directions from sites of the present study, and Fig. 11c shows mean directions for all sites after inverting polarity of directions with negative inclinations. Fig. 11d–f shows the site mean directions calculated for sedimentary rocks. Fig. 11d shows mean directions for six sites (far from any sill) that were attributed to the Vale da Promissão Formation by D'Agrella-Filho et al. (2008). Now these sediments are reinterpreted as Rio Branco sedimentary

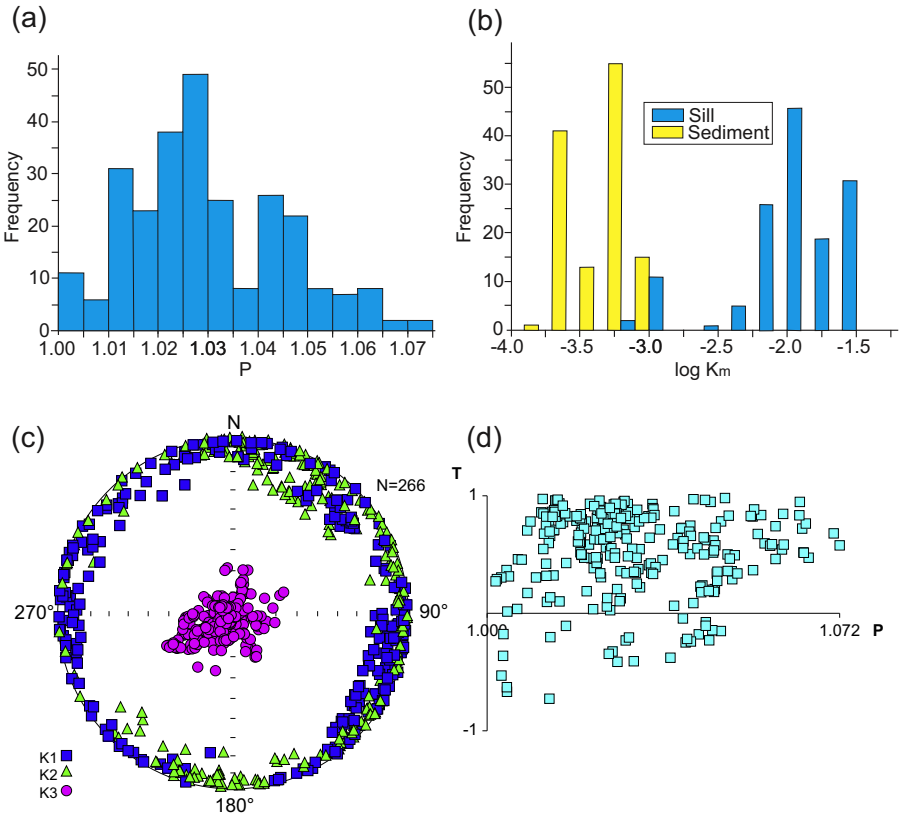


Fig. 6. (a) Histograms showing anisotropy degree (P) frequency; (b) logarithm of mean magnetic susceptibility (log K_m) frequency; (c) directions of the maximum (K_1), intermediate (K_2) and minimum (K_3) susceptibility axes for all analyzed samples; (d) Jelinek parameter (T) versus anisotropy degree (P).

rocks. Fig. 11e includes mean directions from sites of the present study, and Fig. 11f shows mean directions for all sites inverting polarity of site AZ400/401.

A new paleomagnetic pole was calculated for the 18 sites from Salto do Céu Intrusive Suite, which is situated at 56.0°S; 278.6°E (A_{95} = 7.9°, K = 20) (Table 1). This pole supersedes the previous pole published by Elming et al. (2009a) since statistics is improved by the larger number of entries used in the mean. We also decided to rename the pole as Salto do Céu paleomagnetic pole to match the name of the corresponding intrusive suite and to avoid confusion with the much younger Aguapeí Group. A new paleomagnetic pole was also calculated for the sedimentary rocks combining the data of D'Agrella-Filho et al. (2008) with the eight sites studied here (Table 2) which is located at 45.5°S; 270.0°E (A_{95} = 6.5°, K = 39) and

Table 2
Paleomagnetic data for the Rio Branco sedimentary rocks.

Site	Samples	Glat/Glong	Site mean directions						VGP	
		(°S/°E)	n	Dec. (°)	Inc. (°)	α_{95} (°)	K	N, R	Plat. (°N)	Plong. (°E)
20[a]		15.402/301.901	10	195.1	50.4	5.8	69	R	−69.1	263.3
21[a]		15.400/301.901	10	215.8	53.6	7.9	38	R	−52.8	248.8
22[a]		15.300/301.956	9	232.5	67.6	12.2	25	R	−35.2	263.8
23[a]		15.131/301.875	20	221.1	66.9	4.7	49	R	−42.1	266.8
24[a]		15.131/301.875	15	209.1	70.8	7.2	29	R	−44.1	279.1
25[a]		15.131/301.875	16	192.0	70.3	3.0	150	R	−49.7	291.0
AZ252[b]	25 (A-Z)	15.064/301.867	19	244.1	76.0	5.6	37	R	−24.9	275.6
AZ254[b]	3 (A-C)	15.183/301.936	4	242.0	72.9	9.9	88	R	−27.3	270.5
AZ200[b]	15 (A-H; S-Z)	15.054/301.862	21	217.7	62.6	3.3	94	R	−46.4	262.0
AZ201[b]	11 (A-K)	15.054/301.862	18	222.0	64.3	2.5	195	R	−43.2	262.7
AZ225[b]	8 (A-H)	15.169/301.906	17	209.0	65.8	1.4	654	R	−49.3	272.1
AZ226[b]	7 (A-G)	15.169/301.906	9	218.3	65.8	6.3	69	R	−44.4	266.7
AZ227[b]	14 (A-N)	15.169/301.906	22	206.0	68.6	1.9	276	R	−47.7	278.4
AZ400/401[b]	4 (A-D) 1 block	15.354/301.916	10	20.7	-63.6	6.2	61	N	55.4	275.9
Mean R			13	215.8	66.5	4.8	76	R	−44.7 A_{95} = 6.8°	269.6 K = 38
Mean $N+R$			14	214.1	66.2	4.5	78	R, N	−45.5 A_{95} = 6.5°	270.0 K = 39

Glat/Glong – geographic latitude and longitude of the sampled sites; n – number of specimens used in the mean; Dec. – declination; Inc. – inclination; α_{95} – radius of the 95% cone of confidence; K – Fisher precision parameter; VGP – virtual geomagnetic pole; Plat. – pole latitude; Plong. – pole longitude.

[a] From D'Agrella-Filho et al. (2008).

[b] This work.

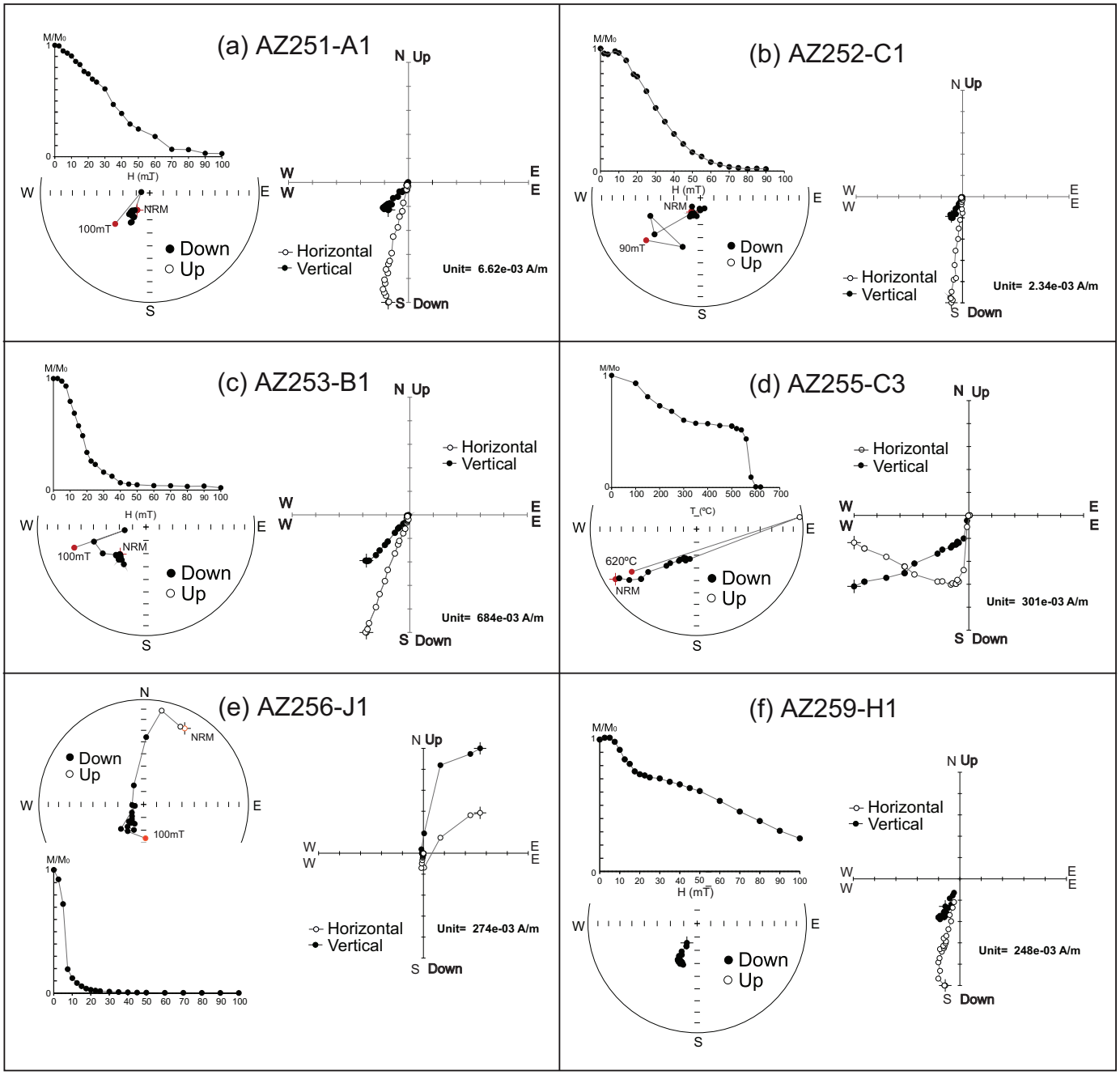


Fig. 7. Examples of AF and thermal demagnetization for samples from the mafic sills. Stereographic projection, normalized (M/Mo) magnetization intensity versus alternating field (AF) or temperature (T) curve, and orthogonal (Zijderveld) horizontal and vertical projections are shown for each sample.

also supersedes the one reported by D'Agrella-Filho et al. (2008). We have also renamed this pole as Rio Branco sedimentary rocks pole, since the new age constraints invalidate their correlation with the Neoproterozoic Vale da Promissão Formation.

6.2. Baked contact test

A positive baked contact test requires that similar directions are isolated for the intrusion and its baked zone, and that a different and more ancient direction is isolated for rocks sampled far from the contact (e.g., Buchan, 2007; Elming et al., 2009b). Thus, the baked contact tests performed for five sills in the present work failed to prove the primary origin of their ChRM direction since directions disclosed for host rocks are always similar regardless of the distance from the sills (Figs. 8–11). This ChRM direction is also similar to that found for other Salto do Céu sills.

Although mean directions calculated for sedimentary rocks and sills are similar (Table 1), a statistical test (McFadden and Lowes, 1981) for discrimination of these mean directions must be applied to ascertain at what level of probability they are distinct. Considering two populations ($N1$ and $N2$) with Fisher distributions (Fisher, 1953) and similar precisions (κ_1 and κ_2), the hyphothesis they share a common true mean direction may be rejected at the level of confidence P, if:

$$\frac{R1 + R2 - (R^2/(R1 + R2))}{2(N - R1 - R2)} > \left(\frac{1}{P}\right)^{1/(N-2)} - 1, \quad (1)$$

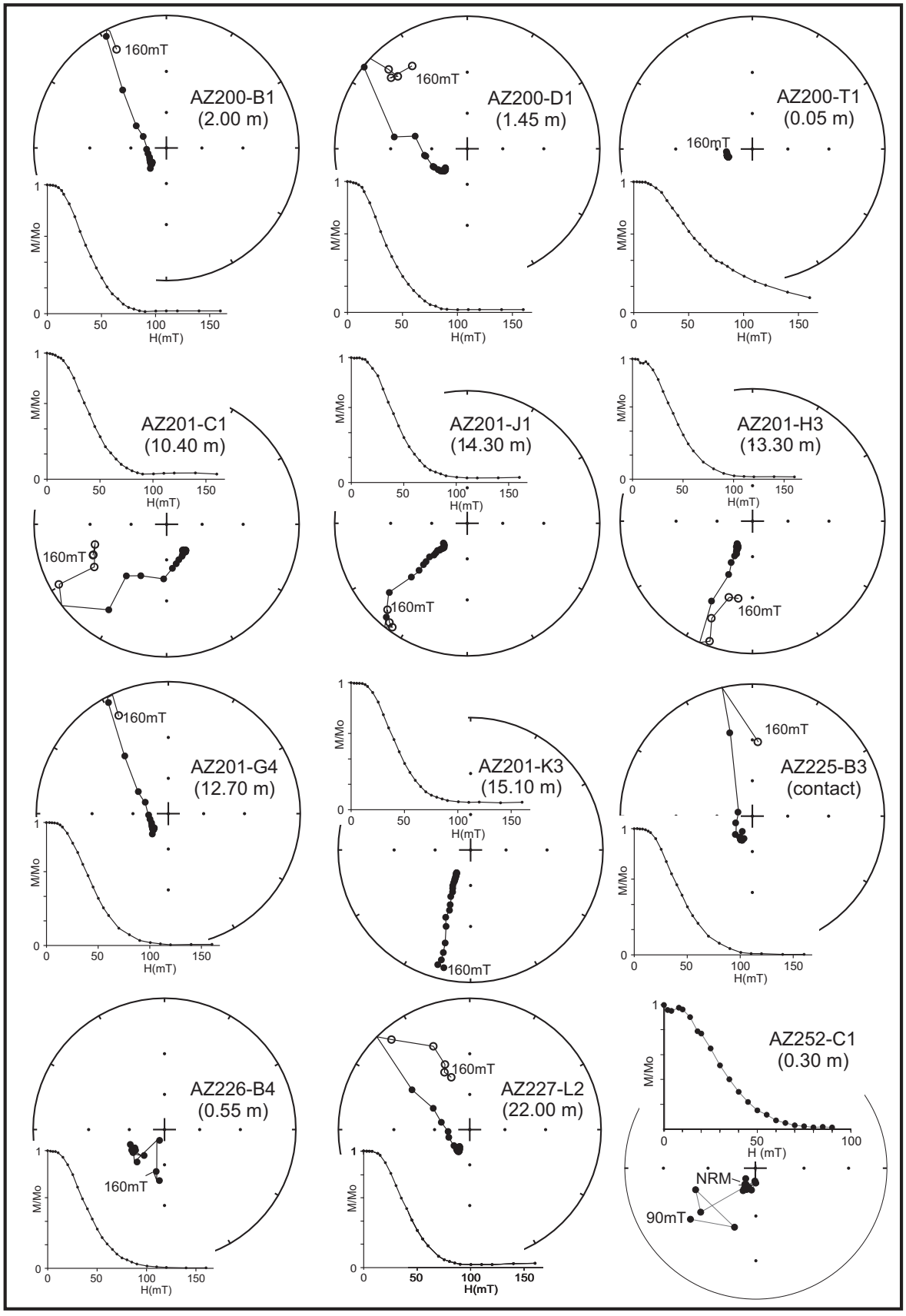


Fig. 8. Examples of AF demagnetization for sedimentary rock samples collected at different distances (shown in meters between parentheses) from the mafic sills. Stereographic projection and normalized (*M*/Mo) magnetization intensity versus alternating field (AF) curve are shown for each sample.

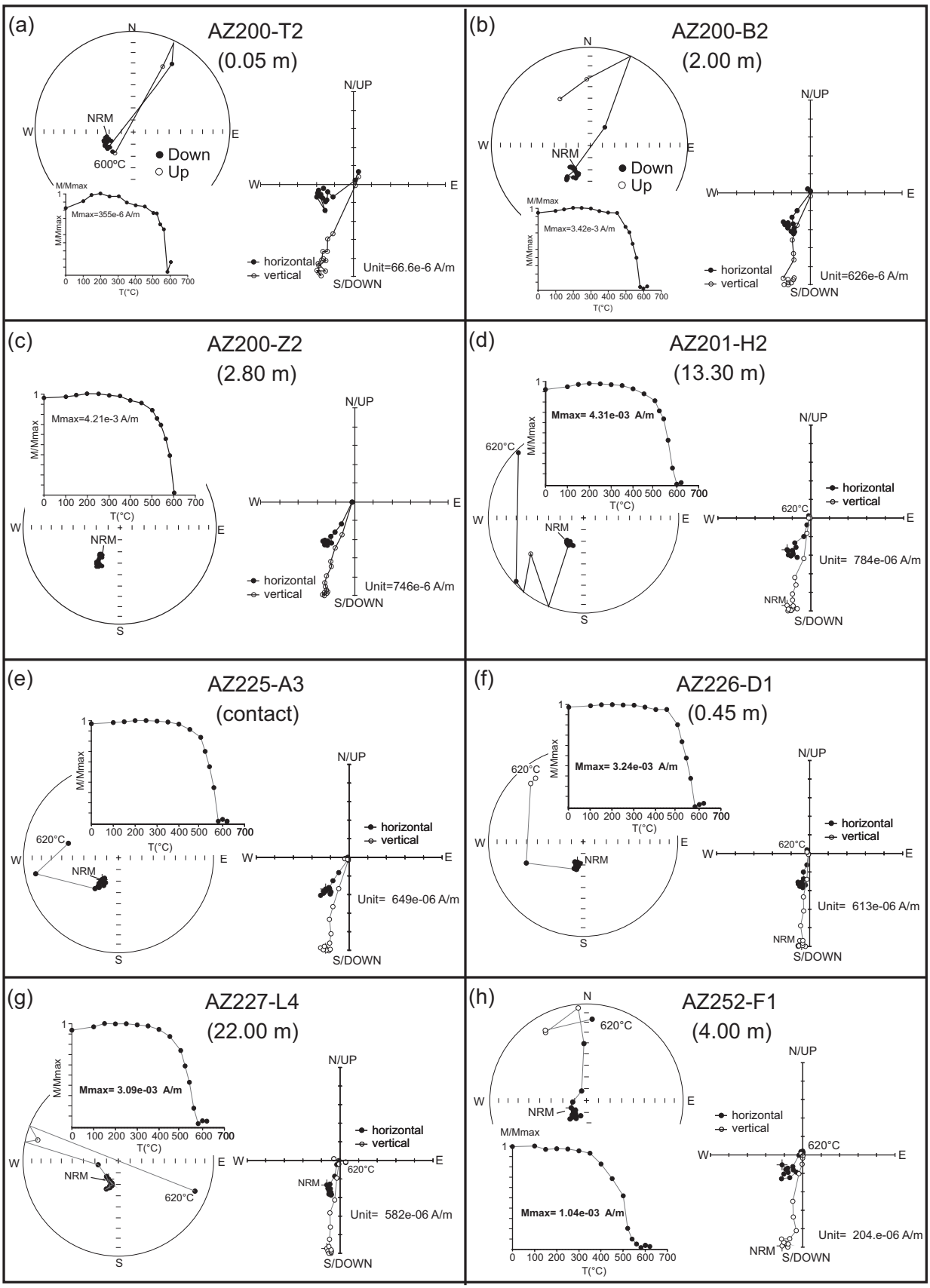


Fig. 9. Examples of thermal demagnetization for sedimentary samples collected at different distances (shown in meters between parentheses) from the mafic sills. Stereographic projection, normalized (M/Mo) magnetization intensity versus temperature (T) curve, and orthogonal (Zijderveld) horizontal and vertical projections are shown for each sample.

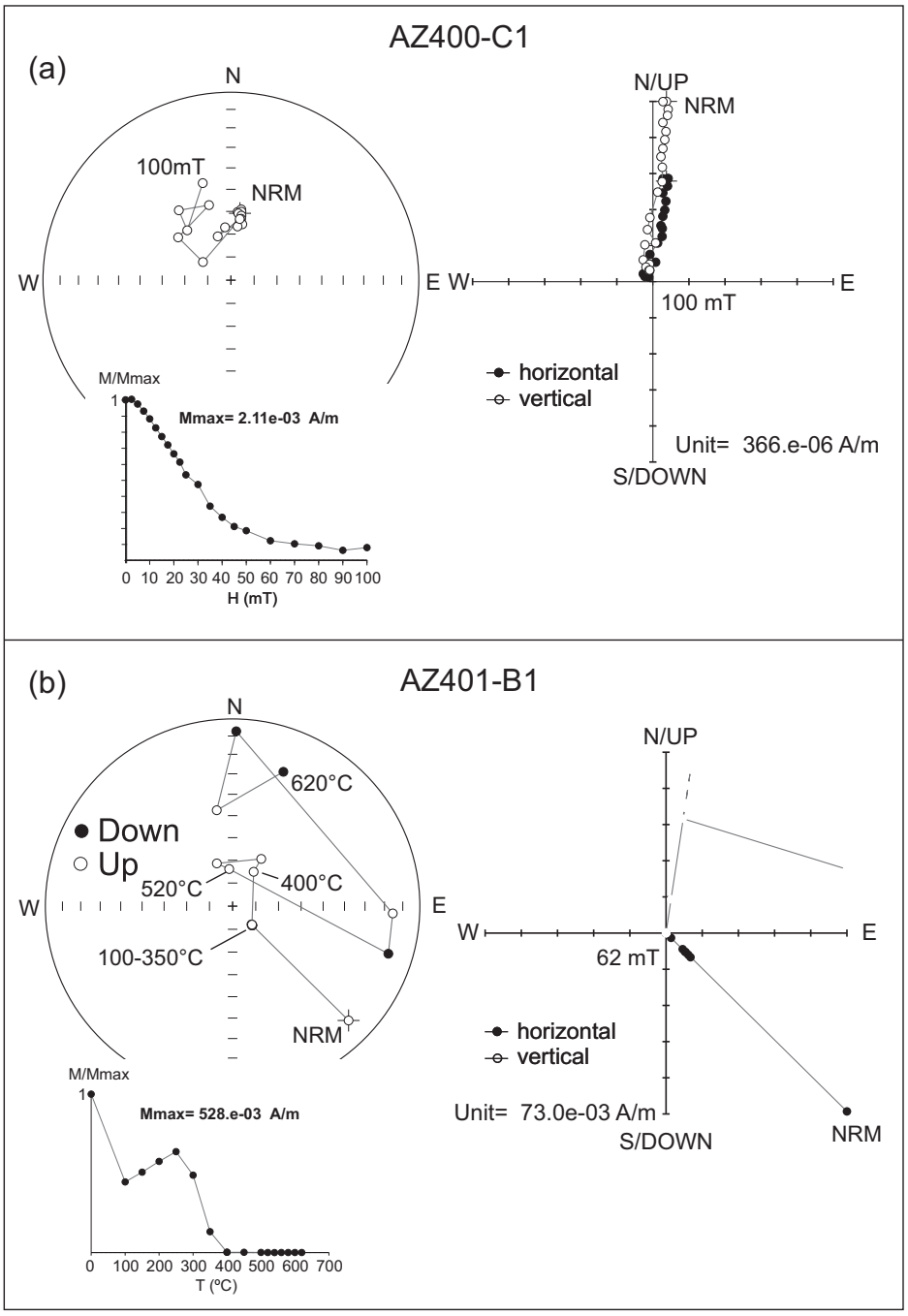


Fig. 10. Examples of AF (a) and thermal (b) demagnetization for sedimentary samples from site AZ400/401, collected close to a sill (site 11 of Elming et al., 2009a). Distance from the sill is not available (see text for details). Stereographic projection, normalized (M/Mo) magnetization intensity versus alternating field (AF) or temperature (T) curve, and orthogonal (Zijderveld) horizontal and vertical projections are shown for each sample.

where $N = N1 + N2$, and R is the length of the vector sum of resultants $R1$ and $R2$ of the separate populations. κ_1 and κ_2 are estimated from Fisher' statistics ($\kappa_i = (Ni - 1)/(Ni - Ri)$).

For populations with different precision parameters, the hyphothesis of a common mean direction is rejected if:

$$\frac{r[(R1+R2)^2 - R^2]}{\{2[(N1-R1)+r(N2-R2)][R1+r+R2]\}} > \left(\frac{1}{P}\right)^{1/(N-2)} - 1, \quad (2)$$

where $r = \kappa_2/\kappa_1$.

Tables 1 and 2 show that precision parameters of mean directions calculated for sills and sedimetary rocks are very different. So, Eq. (2) must be used to test if the null hypothesis that they were drawn from Fisher distributions with the same mean direction can be rejected. Table 3 shows that mean directions for sedimentary rocks and sills are statistically different from each other at the 95% probability level. The same test also shows they are statistically different at 98% probability level.

Given these results, a later metamorphic event that would have affected both sills and sedimentary rocks seems to be unlikely. If this was the case, a common mean between sills and sediments would be expected. In addition, stepwise demagnetization and magnetic mineralogy studies show the ChRM directions on both sills and sedimentary rocks are carried by high-coercivity SD-magnetites with high unblocking temperatures (>500° C). Such

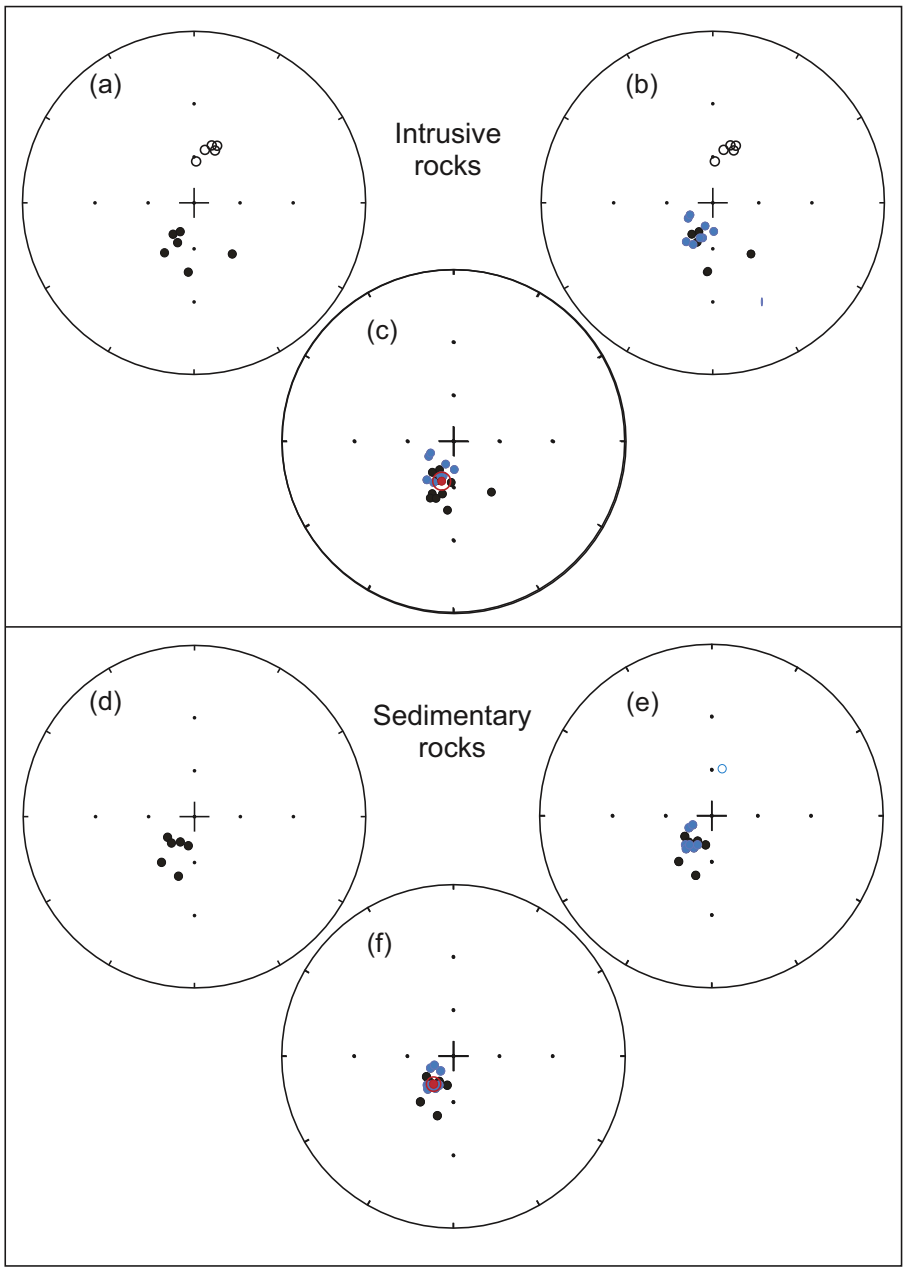


Fig. 11. Site mean directions (a) for the sills and dykes analyzed by Elming et al. (2009a); (b) including the sills from this study (in blue); (c) for all sills reversing the 'normal' polarities (the mean direction with its confidence circle are shown in red); (d) for the sedimentary rock sites analyzed by D'Agrella-Filho et al. (2008); (e) including the sedimentary rock sites from this study (in blue); and for all sedimentary rocks reversing the 'normal' polarity (the mean direction with its confidence circle are shown in red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

high blocking temperatures would only be unblocked if rocks were submitted to regional peak temperatures above 450 °C for several tens or even hundreds of millions of years (Pullaiah et al., 1975). K–Ar and ^{40}Ar–^{39}Ar radiometric analyses on plagioclase and whole rock of sills yielded ages between 1035 ± 3 Ma and 875 ± 21 Ma (Barros et al., 1982; Ruiz, 1992, 2005; Elming et al., 2009a) suggesting they were thermally affected by the Aguapeí orogenic event (to the west of the Rio Branco area – Fig. 1). However, hornblende and biotite ^{40}Ar–^{39}Ar dating of basement rocks preserves igneous crystalline ages, excluding high temperature resetting in the region

Table 3
Comparison of Salto do Céu and Rio Branco sedimentary rocks mean directions.

Pole 1/$N1$	Pole 2/$N2$	$R1$	$R2$	R	N	$R=\kappa_2/\kappa_1$	U	V	$(1-P)\times 100$ (%)
Sills/18	Sed./13	17.55	12.84	30.31	31	2.0193	0.1459	0.1088	98

$U = \frac{r[(R1+R2)^2-R^2]}{|2[(N1-R1)+r(N2-R2)](R1+rR2)|}$

$V = \left(\frac{1}{P}\right)^{1/(N-2)} - 1$

The last column presents the probability of distinctness ($V=U$).

after emplacement of sills (Tohver et al., 2006). This interpretation is further corroborated by petrographic evidence showing that metamorphism is restricted to the sedimentary rocks and only attained low greeschist facies (sericite, chlorite and magnetite overgrowths), which imply these rocks were heated to temperatures not higher than 300 °C. The lack of deformation in sediments and sills as attested by petrographic examination and also from the dominantly planar AMS fabric and their low anisotropy degrees (<1.07) (Fig. 2) suggest that heating of sediments likely resulted from the intrusion of the sills. In addition, it is worth noting that paleomagnetic poles calculated for Salto do Céu sills and sedimentary rocks do not match any available late Neoproterozoic or Phanerozoic pole from South America (Trindade et al., 2006; Font et al., 2011).

We favor a scenario in which sills have intruded a recently deposited sedimentary cover, bringing heat and warm fluids, sometimes causing resetting of the paleomagnetic record of the host porous sediments. Temperature in the surroundings of a magmatic intrusion depend on several factors, including the width of the sills, the intrusion depth, initial temperature of the magma, and temperature of host rock at the moment of intrusion. Also critical are cooling rates of the sills, as well as the variation of thermal diffusivity and conductivity with temperature (Jaeger, 1964; Delaney, 1987; Nabelek et al., 2012).

Regarding all these variants, Nabelek et al. (2012) proposed a model for a 50 m thick basic sill with initial temperature of 1230 °C, intruding into rocks with diffusivity of the mean crust, at about 250 m deep. In their model, host rock's temperature close to the contact (up to 5 m) slightly exceeds ~700 °C but decreases to 500 °C and 300 °C respectively at 30 m and 100 m away from the contact. In this way, heating at the border of the sills would be enough to reset the magnetization of sediments, but magnetization far from the sills must have been preserved. But mineral transformation in the host rocks can also be induced by hydrothermal circulation due to magma heating, sometimes far from the heat source depending on the permeability of the host rocks, and new magnetic minerals can be formed that would carry a younger magnetization. In the Salto do Céu region, petrographic evidence for magnetite neoformation in the sediments (see above) suggests that its magnetization was affected by the sills but detrital magnetite must have preserved a primary signal too. If the age of intrusion is close to the age of sedimentation magnetic directions recorded by both lithologies are expected to be similar as observed.

6.3. Age of sedimentary rocks and Rio Branco sedimentary rocks pole

The recent 1439 ± 4 Ma U–Pb (baddeleyite) age (Teixeira et al., 2015) obtained for the Salto do Céu sill establishes a lower limit of the pelitic sediment deposition in the Rio Branco area. Also, recently Geraldes et al. (2014) presented a provenance study on 100 detrital zircons extracted from these sedimentary rocks in the Salto do Céu region, and they interpret the younger population age (peak of 1544 Ma) as the maximum age of sediment deposition. These ages constrain the sedimentation age to the time interval 1544–1440 Ma. The fact that paleomagnetic poles for both units are similar but not statistically the same suggests that the age of the sediments is closer to the age of the sills. An isolated zircon grain shows a $^{206}Pb/^{207}Pb$ apparent age of 1442 ± 35 Ma (see Table 1, sample AG-1 of Geraldes et al., 2014). This age is within the error of the baddeleyite age obtained for the sill itself although it may be related to the heating effect of the sills.

6.4. Mesoproterozoic poles for the Amazonian Craton

The statistical test of McFadden and Lowes (1981) applied to the mean directions of sills and sedimentary rocks they intrude suggests they register the geomagnetic field at about 1.44 Ga. Recently, two Mesoproterozoic poles were obtained for the Nova Guarita dykes (Bispo-Santos et al., 2012) and for the Indiavaí mafic intrusive (D'Agrella-Filho et al., 2012). The Nova Guarita pole was obtained for dykes cropping out ca. 600 km to the northeast of Salto do Céu sills, whose age of 1419 ± 3 Ma is the mean of four well-defined $^{40}Ar–^{39}Ar$ plateaus determined for different dykes. A positive baked contact test attests to the primary nature of ChRM directions isolated for the dykes (Bispo-Santos et al., 2012). The Indiavaí pole was associated to the age of 1416 ± 7 Ma obtained by U–Pb (in zircon) determinations in the same rocks (D'Agrella-Filho et al., 2012, Teixeira et al., 2011). Fig. 12a shows the paleomagnetic poles for Rio Branco sedimentary rocks (A1), Salto do Céu (A2), Nova Guarita (A3) and Indiavaí (A4). The proximity of all these poles suggests similar ages for the magnetization acquired by the four units.

6.5. Paleogeography of Amazonian Craton in Nuna

The SAMBA connection with West Africa, Amazonia and Baltica as a single unit in the Nuna supercontinent was proposed by Johansson (2009). This assembly also incorporates Laurentia, which is classically placed against Baltica in Paleoproterozoic reconstructions (see discussion of Johansson, 2014). Paleomagnetic data favor a connection of Eastern Greenland in Laurentia with the Timanide margin of Baltica, which lasted for hundreds of millions of years, probably up to 1270 Ma, suggesting a long duration for at least this sector of Nuna (Buchan et al., 2000; Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Pisarevsky and Bylund, 2010; Evans and Mitchell, 2011; Pesonen et al., 2012; Zhang et al., 2012; Evans, 2013; Bispo-Santos et al., 2014a).

In Fig. 12a, the 1440–1420 Ma Amazonian poles are compared with Mesoproterozoic poles from Baltica and Laurentia after their rotations to the SAMBA configuration, i.e. with southern Baltica facing northwest Amazonia (Bispo-Santos et al., 2014a). The Amazonian Mesoproterozoic poles plot away from those of Baltica and Laurentia suggesting a demise of the SAMBA connection by that time. Note, however, that all the Amazonian poles are from geological units situated in the Brazil-Central (or Guaporé) shield in the southern part of the Amazonian Craton (Fig. 12a). Interestingly, the 1790–1780 Ma Colider pole (Bispo-Santos et al., 2008), also from the southeastern part of Amazonian Craton, is situated to the West of the ~1790 Ma paleomagnetic pole of Avanavero dykes that are from the northern Amazonian Craton (see Fig. 12 of Bispo-Santos et al., 2014a). An interpretation for the differences in both Paleo- and Mesoproterozoic poles from southern Amazonia and northern Amazonia/Baltica/Laurentia is that dextral relative motions occurred between the Brazil-Central shield and the rest of Nuna or the partition of the movement within the central portion of the craton. Bispo-Santos et al. (2014a) discuss possible tectonic processes that could be responsible for these relative movements. Pisarevsky et al. (2014) also discuss the Paleoproterozoic (Colider and Avanavero) and Mesoproterozoic (Nova Guarita and Indiavaí) poles from Amazonia, and possible displacements between northern and southern Amazonia. They admit the possibility of the SAMBA model based on the Avanavero pole, but argue that displacements between parts of Amazonia are unlikely, as they would disrupt the linearity of the Ventuary-Tapajós province. Since geological and paleomagnetic evidence suggest that Baltica was not formed at 1790 Ma, Pisarevsky et al. (2014) preferred to position Amazonia/West Africa outboard of the peripheral subduction system comprised by Laurentia and Baltica at 1770 Ma. According to Pehrsson et al. (2015), however, this would bring a greater difficulty to the paleogeographic evolution from Nuna to Rodinia (see also Evans, 2013).

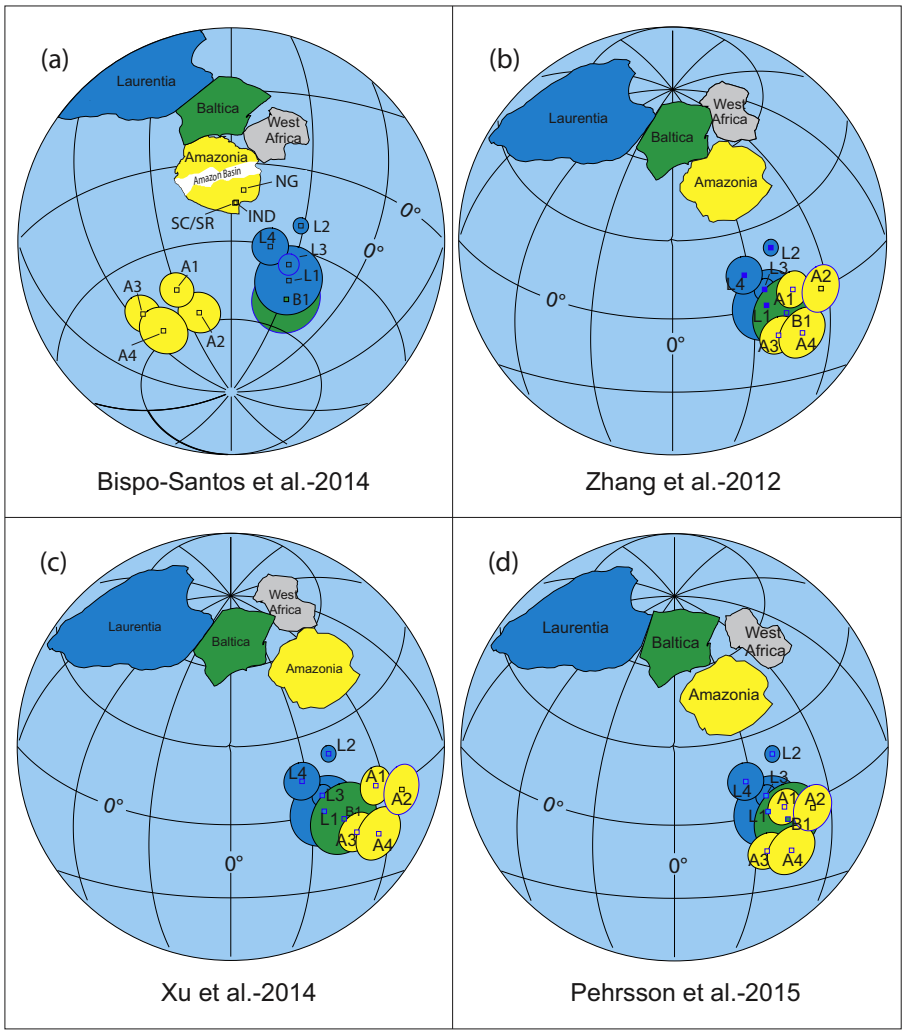


Fig. 12. Comparison of Mesoproterozoic poles from Amazonian Craton, Laurentia, and Baltica considering the reconstruction of Nuna proposed by (a) Bispo-Santos et al. (2014a): Amazonian Craton and respective poles in their present position. West Africa was rotated −71.5° around the Euler pole at 43.3°N; 330.5°E (according to Bispo-Santos et al., 2014b). Baltica and respective pole were rotated −78.7° around the Euler pole at 43.4°N; 200.3°E (Bispo-Santos et al., 2014a). Laurentia and respective poles were rotated −84.0° around the Euler pole at 54.2°N; 249.2°E (Bispo-Santos et al., 2014a); (b) Zhang et al. (2012): Laurentia and respective poles in their present position. West Africa was rotated −70.7° around the Euler pole at 30°N; 266°E; Baltica and respective pole were rotated 49° around the Euler pole at 47.5°N; 1.5°E. Amazonia and respective poles were rotated 127° around the Euler pole at 53°N, 293°E; (c) Xu et al. (2014): Laurentia and respective poles in their present position. West Africa was rotated to Amazonia according to Bispo-Santos et al. (2014b); Baltica and respective pole were rotated 49° around the Euler pole at 47.5°N; 1.5°E. Amazonia (plus West Africa) and respective poles were rotated 138° around the Euler pole at 58°N, 299.9°E; (d) Pehrsson et al. (2015): Laurentia and respective poles in their present position. West Africa was rotated 85.8° around the Euler pole at 58.3°N; 304.9°E; Baltica and respective pole were rotated 49° around the Euler pole at 47.5°N; 1.5°E. Amazonia and respective poles were rotated 122.2° around the Euler pole at 53.9°N, 291.1°E. Paleomagnetic poles: Amazonia – (A1) Rio Branco Sedimentary Rocks (this work), (A2) Salto do Céu intrusives (this work), (A3) Nova Guarita Dykes (Bispo-Santos et al., 2012) and (A4) Indiaval Intrusive (D'Agrella-Filho et al., 2012); Baltica: (B1) 1460 Ma mean pole: Plat. = 15.0°S; Plong. = 0.0°E; A_{95} = 13.0° (Bispo-Santos et al., 2014a); Laurentia: (L1) 1460 Ma mean pole: Plat. = 1.0°N; Plong. = 35.0°E; A_{95} = 13.0°, (L2) McNamara pole (1401 ± 6 Ma): Plat. = 14.0°N; Plong. = 28.0°E; A_{95} = 7.0°, (L3) Electra Lake Gabbro (1433 ± 2 Ma): Plat. = 21.0°N; Plong. = 41.0°E; A_{95} = 3.0°, (L4) Laramie Anorthosite (1429 ± 9 Ma): Plat. = 7.0°N; Plong. = 35.0°E; A_{95} = 4.0° (Bispo-Santos et al., 2014a). Paleomagnetic poles are represented in the same color of the respective continental blocks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Recently other models of Nuna have been proposed (e.g., Zhang et al., 2012; Xu et al., 2014; Pehrsson et al., 2015) for which smaller mismatches of the Mesoproterozoic poles from Amazonia, Baltica and Laurentia are observed. Fig. 12 shows reconstructions proposed by Zhang et al., 2012 (Fig. 12b), Xu et al., 2014 (Fig. 12c) and Pehrsson et al., 2015 (Fig. 12d). For each reconstruction selected 1460–1400 Ma poles from Baltica and Laurentia are compared with the four Mesoproterozoic poles from Amazonia. The best cluster of poles is obtained with the reconstruction of Pehrsson et al. (2015), where the Amazonia appears rotated counterclockwise relative to the reconstruction of Bispo-Santos et al. (2014a) (Fig. 12a), and may indicate internal plate rotations inside Nuna. Note that this reconstruction is similar to that proposed by Bispo-Santos et al. (2012). However, new Mesoproterozoic poles from the Amazonian Craton, mainly from the northern Guiana Shield, are required before we decide in favor of one of the models proposed for Nuna.

7. Conclusions

- A more robust paleomagnetic pole was calculated from 18 sites of the Salto do Céu Intrusive Suite, which is located at 56.0°S; 278.6°E (A_{95} = 7.9°, K = 20). This pole integrates previous results and supersedes the Aguapeí pole (Elming et al., 2009a). The pole was renamed as Salto do Céu after the original regional name of the intrusives.
- A combined paleomagnetic pole was calculated for the sedimentary rocks (45.5°S; 270.0°E, N = 14, A_{95} = 6.5°, K = 39) which was renamed the Rio Branco sedimentary rocks pole, since the new

age constraints are not compatible with previous correlations of these rocks with the Neoproterozoic Vale da Promissão Formation.

- The new geochronological data invalidates the use of the Aguapeí pole as a reference pole to reconstruct the position of the Amazonian Craton in Rodinia. Since that was the only Amazonian reference pole for the Tonian to Cryogenian periods, new data is thus needed to constrain the final position of the Amazonian Craton in Rodinia.
- The new data match previously published Mesoproterozoic results for Amazonia and are compatible with a connection between Amazonia and Baltica at the Mesoproterozoic in a paleogeographic configuration slightly different from SAMBA (South America-Baltica).

Acknowledgements

M.S. D'Agrella-Filho and F. Bispo-Santos thank the financial support from Fundo de Amparo à Pesquisa do Estado de São Paulo (FAPESP – grant 2011/50887-6 and 2012/50327-3) and Conselho Nacional de Pesquisa (CNPq – grant 303130/2014-8). This research was performed as part of the INCT-GEOCIAM (Instituto Nacional de Ciência e Tecnologia de Geociências da Amazônia). We thank the reviewers D.A.D. Evans and Å. Johansson and the editor Wilson Teixeira for their comments that greatly improved our paper.

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